water at solid interfaces[35]. However, if the same model is applied to explain the in-plane response [three capacitors in parallel, yielding the effective $\varepsilon_{//}(h) = \varepsilon_{bulk} + 2(\varepsilon_{//int} - \varepsilon_{bulk})h_{int}/h$], the observed $\varepsilon_{//}(h)$ dependence cannot be described even qualitatively (light red dashed curve in Fig. 3a). The sharp rise at $h \approx$ 2-3 nm clearly indicates a transition in the dielectric properties of water if the two confining surfaces get sufficiently close so that the near-surface layers merge and water develops a more distinctly layered structure[4,36]. The large increase in $\varepsilon_{//}$ in this case is consistent with theory[28,29], although the transition was envisaged for $h \gtrsim 1$ nm. Let us also mention the slight tendency seen in Fig. 3a at $h > 4$ nm for $\varepsilon_{//}(h)$ to increase with decreasing h. This increase would be consistent with many predictions[27-30] and, within the 3-layer model and our experimental accuracy, suggests values of $\varepsilon_{//int}$ up to ~150 for interfacial water. Measurements at higher accuracy are required to corroborate the latter observation.

In-plane conductivity

The extracted conductivities for nanoconfined water are plotted in Fig. 3b. One can see that, for moderate confinement, $\sigma_{//}(h)$ evolves approximately as $1/h$ increasing from 10^{-3} - 10^{-4} S·m^{-1} for our largest channels up to ~0.1 S·m^{-1} at $h \approx 3$ nm. This gradual increase is well described by the discussed 3-layer model (inset of Fig. 3a) assuming unaffected conductivity for the inner layer and the presence of two near-surface layers with interfacial conductivity $\sigma_{//int}$, which yields $\sigma_{//}(h) = \sigma_{bulk} + 2(\sigma_{//int} - \sigma_{bulk})h_{int}/h$. Using the same h_{int} for interfacial water as for the dielectric constant, we estimate $\sigma_{//int}$ as ~0.1 S·m^{-1}, three orders of magnitude higher than σ_{bulk} (σ_{bulk} was measured independently using our bulk water with pH ≈ 5.7 caused by CO_2 absorption). Such an enhancement of confined water's conductivity agrees well with previous reports for water inside nanochannels and is attributed to the surface charge[15,16,19,33,34].

For quasi-2D water, where the water becomes layered across its entire thickness ($h \gtrsim 2$ nm), we observe a pronounced additional peak in conductivity. The behavior clearly diverges from the surface charge (3-layer) model but correlates well with the anomalous increase in $\varepsilon_{//}$ (cf. Figs. 3a and 3b). At its peak, $\sigma_{//}$ reaches ~3 S·m^{-1}, which occurs at $h \approx 1.5$ nm and corresponds to 4-5 molecular layers[36]. This value is one of the highest conductivities reported at RT, comparable to that of Nafion and approaching the superionic levels reached for molten salts (~ 10 S·m^{-1})[37]. Note that our smallest channels ($h \approx 1.0$ nm) exhibited a marked (> 10 times) decrease in $\sigma_{//}$ with respect to the peak value. To corroborate the presence of the peak for quasi-2D water, Fig. 3b incorporates a data point available in the literature (grey symbol), which was obtained for water inside 0.7 nm hBN channels using DC measurements[33]. This also shows consistency between different types of experiments.

Discussion

Our experiment reveals that water under confinement exhibits two distinct regimes. Under moderate confinement (h down to ~ 2-3 nm), water can be described as effectively containing inner bulk water and interfacial water. The latter is characterized by enhanced conductivity (by at least 3 orders of magnitude for the case of hBN surfaces) and a highly anisotropic dielectric constant with $\varepsilon_{\perp} \approx 2$ and $\varepsilon_{//}$ comparable to ε_{bulk} or slightly larger. For the extreme, atomic-scale confinement that allows only a few molecular layers inside, the quasi-2D water clearly switches into another state. It is different from either bulk or interfacial water and exhibits ferroelectric-like polarizability and superionic-like conductivity (presumably provided by proton transport).

The electrical behavior observed under the moderate (≥ 3 nm) confinement generally agrees with the previous reports and theoretical expectations[15,19,27-30]. As for the atomic-scale confinement, both measurements[4] and molecular dynamic simulations[36] have previously revealed that water has a distinctive layered structure exhibiting more profound density oscillations than those found for near-surface water. The observed anomalies in the electrical properties of few-layer water have not been anticipated in theory, although giant, ferroelectric-like values of $\varepsilon_{//}$ were suggested in Refs.[28,29] and superionic conductivity was predicted for monolayer water at high temperatures[38]. Furthermore, experiments on water flow and ionic transport also reported the emergence of anomalies in the same few-layer regime[33,39].

In the absence of a theory, we note that atomic-scale confinement induces strong hydrogen-bond disorder because of constraints on water molecules' possible orientations and, also, a loss in hydrogen bonding imposed by the two confining surfaces. Qualitatively, this suggests changes in the electrical properties analogous to those found for disordered hydrogen-bonded crystals[40] and in particular in disordered ice. The latter is known to exhibit higher ε than that of liquid water (and ordered ice) due to easier water molecules' reorientation[14]. Similarly, we speculate that the disruption of the hydrogen-bond network in quasi-2D water enables correlated reorientations of molecular dipoles, allowing their collective polarization[41], and consequently higher, ferroelectric-like $\varepsilon_{//}$[12,28]. Water dipoles' reorientation is essential in the Grotthuss mechanism that explains fast proton transport in water. Accordingly, correlated dipolar reorientations in quasi-2D water should also facilitate faster proton transfer, resulting in higher $\sigma_{//}$, on top of the increase in near-surface proton concentration due to surface charges.

Conclusion

We have measured the in-plane electrical properties of 2D nanoconfined water and found that they radically change as the water's layered structure becomes more pronounced under extreme, molecular-scale confinement. We expect this behavior to be general for strongly confined water and not limited to the case of hBN channels studied here. Our results are fundamental to interpreting numerous physical and biochemical processes occurring at the molecular scale. In particular, they provide insights into the electric double layer under strong confinement, which is essential for developing energy storage and generation technologies. The reported approach to measure conductivities and dielectric constants can be applied to substances other than water and has the notable advantage that it probes local properties without averaging over a distribution of different confinement strengths as inevitable in macroscale measurements.

References

1 Eisenberg, D. & Kauzmann, W. *The structure and properties of water*. (OUP Oxford, 2005).

2 Gonella, G. *et al.* Water at charged interfaces. *Nat. Rev. Chem.* **5**, 466-485 (2021).

3 Muñoz-Santiburcio, D. & Marx, D. Confinement-controlled aqueous chemistry within nanometric slit pores. *Chem. Rev.* **121**, 6293-6320 (2021).

4 Israelachvili, J. N. & Pashley, R. M. Molecular layering of water at surfaces and origin of repulsive hydration forces. *Nature* **306**, 249-250 (1983).

5 Lee, C. Y., McCammon, J. A. & Rossky, P. The structure of liquid water at an extended hydrophobic surface. *J. Chem. Phys.* **80**, 4448-4455 (1984).

6 Israelachvili, J. N. *Intermolecular and surface forces*. (Academic Press, 2011).

7 Nandi, N., Bhattacharyya, K. & Bagchi, B. Dielectric relaxation and solvation dynamics of water in complex chemical and biological systems. *Chem. Rev.* **100**, 2013-2046 (2000).

8 Born, M. Volumen und hydratationswärme der ionen. *Z. Physik* **1**, 45-48 (1920).
9 Breynaert, E. *et al.* Water as a tuneable solvent: a perspective. *Chem. Soc. Rev.* **49**, 2557-2569 (2020).
10 Fumagalli, L., Esteban-Ferrer, D., Cuervo, A., Carrascosa, J. L. & Gomila, G. Label-free identification of single dielectric nanoparticles and viruses with ultraweak polarization forces. *Nat. Mater.* **11**, 808-816 (2012).
11 Pimentel, G. C. & McClellan, A. Hydrogen bonding. *Annu. Rev. Phys. Chem.* **22**, 347-385 (1971).
12 Kirkwood, J. G. The dielectric polarization of polar liquids. *J. Chem. Phys.* **7**, 911-919 (1939).
13 Kreuer, K.-D. Proton conductivity: materials and applications. *Chem. Mater.* **8**, 610-641 (1996).
14 Petrenko, V. F. & Whitworth, R. W. *Physics of ice*. (OUP Oxford, 1999).
15 Stein, D., Kruithof, M. & Dekker, C. Surface-charge-governed ion transport in nanofluidic channels. *Phys. Rev. Lett.* **93**, 035901 (2004).
16 Duan, C. & Majumdar, A. Anomalous ion transport in 2-nm hydrophilic nanochannels. *Nat. Nanotechnol.* **5**, 848-852 (2010).
17 Siria, A. *et al.* Giant osmotic energy conversion measured in a single transmembrane boron nitride nanotube. *Nature* **494**, 455-458 (2013).
18 Sun, P., Ma, R. & Sasaki, T. Recent progress on exploring exceptionally high and anisotropic H+/OH− ion conduction in two-dimensional materials. *Chem. Sci.* **9**, 33-43 (2018).
19 Artemov, V. G. *et al.* Anomalously high proton conduction of interfacial water. *J. Phys. Chem. Lett.* **11**, 3623-3628 (2020).
20 Stern, O. Zur theorie der elektrolytischen doppelschicht *Elektrochem. Angew. Phys. Chem.* **30**, 508-516 (1924).
21 Conway, B. E., Bockris, J. O. M. & Ammar, I. A. The dielectric constant of the solution in the diffuse and Helmholtz double layers at a charged interface in aqueous solution. *Trans. Faraday Soc.* **47**, 756-766 (1951).
22 Fumagalli, L. *et al.* Anomalously low dielectric constant of confined water. *Science* **360**, 1339-1342 (2018).
23 Zhang, C., Gygi, F. & Galli, G. Strongly anisotropic dielectric relaxation of water at the nanoscale. *J. Phys. Chem. Lett.* **4**, 2477-2481 (2013).
24 Schlaich, A., Knapp, E. W. & Netz, R. R. Water dielectric effects in planar confinement. *Phys. Rev. Lett.* **117**, 048001 (2016).
25 Matyushov, D. V. Dielectric susceptibility of water in the interface. *J. Phys. Chem. B* **125**, 8282-8293 (2021).
26 Dufils, T. *et al.* Origin of dielectric polarization suppression in confined water from first principles. *Chem. Sci.* **15**, 516-527 (2024).
27 Loche, P., Ayaz, C., Wolde-Kidan, A., Schlaich, A. & Netz, R. R. Universal and nonuniversal aspects of electrostatics in aqueous nanoconfinement. *J. Phys. Chem. B* **124**, 4365-4371 (2020).
28 Motevaselian, M. H. & Aluru, N. R. Confinement-induced enhancement of parallel dielectric permittivity: super permittivity under extreme confinement. *J. Phys. Chem. Lett.* **11**, 10532-10537 (2020).
29 Ruiz-Barragan, S., Muñoz-Santiburcio, D., Körning, S. & Marx, D. Quantifying anisotropic dielectric response properties of nanoconfined water within graphene slit pores. *Phys. Chem. Chem. Phys.* **22**, 10833-10837 (2020).
30 Hamid, I., Jalali, H., Peeters, F. M. & Neek-Amal, M. Abnormal in-plane permittivity and ferroelectricity of confined water: From sub-nanometer channels to bulk. *J. Chem. Phys.* **154**, 114503 (2021).
31 Borgis, D., Laage, D., Belloni, L. & Jeanmairet, G. Dielectric response of confined water films: Insights from classical DFT. *Chem. Sci.* **14**, 11141-11150 (2023).
32 Mangaud, E., Bocquet, M.-L., Bocquet, L. & Rotenberg, B. Chemisorbed vs physisorbed surface charge and its impact on electrokinetic transport: Carbon vs boron nitride surface. *J. Chem. Phys.* **156** (2022).
33 Esfandiar, A. *et al.* Size effect in ion transport through angstrom-scale slits. *Science* **358**, 511-513 (2017).
34 Qin, S. *et al.* High and stable ionic conductivity in 2D nanofluidic ion channels between boron nitride layers. *J. Am. Chem. Soc.* **139**, 6314-6320 (2017).
35 Tocci, G., Joly, L. & Michaelides, A. Friction of water on graphene and hexagonal boron nitride from ab initio methods: very different slippage despite very similar interface structures. *Nano Lett.* **14**, 6872-6877 (2014).
36 Ruiz-Barragan, S., Muñoz-Santiburcio, D. & Marx, D. Nanoconfined water within graphene slit pores adopts distinct confinement-dependent regimes. *J. Phys. Chem. Lett.* **10**, 329-334 (2019).
37 Hull, S. Superionics: crystal structures and conduction processes. *Rep. Prog. Phys.* **67**, 1233 (2004).
38 Kapil, V. *et al.* The first-principles phase diagram of monolayer nanoconfined water. *Nature* **609**, 512-516 (2022).
39 Radha, B. *et al.* Molecular transport through capillaries made with atomic-scale precision. *Nature* **538**, 222-225 (2016).
40 Baranov, A. I. Crystals with disordered hydrogen-bond networks and superprotonic conductivity. Review. *Crystallogr. Rep.* **48**, 1012-1037 (2003).
41 Pomes, R. & Roux, B. Free energy profiles for H+ conduction along hydrogen-bonded chains of water molecules. *Biophys. J.* **75**, 33-40 (1998).

METHODS

Device fabrication. We fabricated our devices using procedures similar to those described in Ref.[22] A free-standing silicon nitride (SiN) membrane was used as a substrate for our vdW assembly that involved four atomically-flat crystals (Extended Data Fig. 1). We started with etching a rectangular aperture of ~3×20 μm^2 in the SiN membrane. This aperture was required to serve later as a water inlet from a reservoir placed at the back of the wafer. Using the dry transfer method, we transferred a graphite crystal (thickness ~ 10 nm) to seal the aperture. The graphite later served as the bottom electrode. It was etched through from the backside using reactive ion etching, which projected the aperture into graphite. Then a relatively thick hBN crystal ($H \approx 50$ - 200 nm; to serve as the bottom layer) was transferred on top of the graphite and again dry-etched through from the back. Next, we selected a second hBN crystal (thickness $h \approx 1$ - 60 nm), referred to as spacer, and patterned it into parallel stripes (spaced by ~ 200 nm) using e-beam lithography and reactive ion etching. We transferred the spacer layer on top of the bottom hBN using wet transfer procedures and aligning the stripes perpendicular to the rectangular aperture. Finally, the third hBN crystal was transferred on top of the spacer layer using the dry transfer method. This sealed the nanochannels as well as the aperture. The thickness of the top hBN ($H_{top} \approx 20$ - 80 nm) was carefully chosen in order to bring the AFM tip as close as possible to the water but also to ensure that the top layer exhibits some sagging into empty channels without blocking them completely[22] (Extended Data Figs. 2a and 2d). After each transfer, we annealed the assembly in Ar/H_2 at 300°C for 3 hours and then at 400°C for 5 hours to remove polymer residues and other contamination. As the final step, we made an electrical contact with the graphite using photolithography or, alternatively, using silver paint to minimize the number of cleanroom processes. Because the top layer became flattened (no longer sagged)[22] after liquid water filled the channels (see Extended Data Figs. 2d-g), this allowed us to ensure that, irrespective of water's dielectric response, the probed nanochannels were fully filled with water (by comparing AFM topography images before and after filling; Extended Data Figs. 2e and 2f). Note that, given the large dielectric response of water found in this work for all channel's thicknesses, we could also verify the water filling by taking AFM images in the intermittent-contact attractive mode using low-f applied voltages (a few kHz down to DC). In this case, the acquired contrast over the channels reversed from negative to positive upon filling the water (Extended Data Figs. 2b and 2c) due to the onset of a strong electrostatic force associated with water's conductivity.

As another improvement with respect to our previous studies, before filling with water, we normally exposed the bottom side of the devices to low-power Ar/O_2 plasma (8 W, 16 sccm Ar and 8 sccm O_2 flow) for 1 sec. This made SiN more hydrophilic and also cleared channels' entrances from possible contamination. This procedure has proven beneficial for getting water inside. We also found that our devices tended to delaminate if contamination remained trapped between vdW layers. To prevent this from happening, great care was taken to check for cleanliness of the devices at each fabrication step (using optical and AFM imaging). For example, if dark-field optical images showed bubbles trapped between layers, such devices were discarded. Representative optical images of the studied devices with clean interfaces are provided in Extended Data Figs. 1a-c.

Local broadband dielectric imaging and spectroscopy. All SDM[10,42] measurements were carried out using a commercial AFM (Nanotec Electronica with WSxM software[43]) operated at RT in dry atmosphere. SDM was implemented in the amplitude-modulated electrostatic-force detection mode[44,45], adapting the approach described in Ref.[10] Briefly, we applied an AC voltage between the AFM tip and the bottom electrode. By detecting mechanical oscillations of the cantilever, we measured the electrostatic force and, therefore, the first derivative of the tip-sample capacitance, dC/dz. Its value depends on both dielectric and conductive properties of probed samples. SDM has previously been used for local measurements of both surface and sub-surface dielectric properties of various materials in different frequency regimes, from quasi-static to GHz[46-49]. The same approach has also been widely used to study near-surface electron transport in solid samples[50-53], showing the technique's sensitivity to local conductivity. The force-sensing approach was preferable in our case to the other current- and microwave-sensing AFM techniques that can also probe local impedance[54,55] because the latter are generally less sensitive and more complex to implement.

To carry out dielectric spectroscopy over the required very wide bandwidth (100 Hz – 1 GHz), we combined two previously reported force-sensing detection methods. Both were implemented here using conductive diamond-coated AFM probes (CDT-CONTR or CDT-FMR from Nanosensors™) with spring constant k of few Nm^{-1} and resonance frequency in the range of 20-60 kHz. At low f up to the cantilever resonance frequency, we used the 2ω-detection approach, as described in Ref.[22] for measurements of water's $\varepsilon_\perp$, where $\omega = 2\pi f$ is the angular frequency. Briefly, we applied an AC voltage with amplitude $v_{AC} = 4$ V and measured the amplitude $D_{2\omega}(z)$ of the resulting mechanical oscillations of the cantilever at double the applied frequency, using an external lock-in amplifier (Zurich Instruments HF2LI). The AFM tip-sample capacitance gradient was calculated as $|dC/dz| = D_{2\omega}(z) \cdot 4k/v_{AC}^2$. For f higher than the cantilever resonance frequency, we used the heterodyne detection technique demonstrated in Ref.[56] To this end, we applied a high-f (carrier) signal modulated by low frequency ($f_{mod} = 1$ kHz, amplitude $v_{AC} = 0.5$ V) using an external RF/GHz signal generator (Rohde & Schwarz SMA100B). We detected the cantilever mechanical oscillations at f_{mod} using our external lock-in amplifier and, from those measurements, obtained the capacitance gradient at the carrier f as $|dC/dz| = D_{mod}(z) \cdot 8k/(g \cdot v_{AC}^2)$, where g is the f-dependent gain of the external circuit, $D_{mod}(z)$ is the amplitude of the cantilever mechanical oscillations at f_{mod}. This approach allowed retrieval of $|dC/dz|$ variations at f higher than the cantilever resonance frequency and up to GHz frequencies[52]. Note that such measurements yield only the amplitude of dC/dz and not its phase, so that the obtained spectra reflect the modulus of the complex dielectric constant (see "Analysis of local dielectric spectra").

To minimize systematic errors, for each device and each AFM tip, we carried out careful calibration of the electrical gain g at each measurement frequency. This is essential especially at high frequencies (> 1 MHz), as both applied and measured signals are subject to large f-dependent variations because the cable impedance is not matched to the local sample impedance and the long-range impedance between the AFM tip and the sample. The gain factor was calibrated using a similar procedure to that introduced in Ref.[48], which relies on acquiring $|dC/dz|$ curves as a function of the tip-surface distance z over a region of the device that exhibits f-independent impedance. We then determined g by comparing $|dC/dz|(z)$ curves at specific frequencies against a low-f (reference) curve, where no gain or loss is expected. To minimize possible changes in the gain in different regions of the device, we took the $|dC/dz|(z)$ curves at various f over the center of the hBN spacer close to the studied water-filled channel (Extended Data Fig. 3a). We then scaled them to match the curve taken at 2 kHz using the gain g and the offset as two fitting parameters (Extended Data Fig. 3c). The latter is always present in dC/dz curves, irrespective of frequency, and is typically subtracted before analysis, because the offset does not depend on local electrical properties (see, e.g., Refs.[10,57]). Indeed, the measured offsets were found to change substantially at high f (10 MHz – 1GHz), but they were the same above water channels and hBN spacers. This confirmed that, for all the frequencies, the offsets originated from long-range impedance contributions. Importantly, while g is critical for accurately evaluating the electrical properties of water from the measured signals, offsets play little role. This is because we analyzed relative changes in $|dC/dz|$ along the device surface, that is, water channels vs hBN spacers (see "Dielectric images and dielectric spectra acquisition"). We verified our calibration procedure using reference samples (hBN on doped Si) and obtained good agreement with the expected f-independent dielectric spectra up to GHz frequencies (Extended Data Fig. 3e,f). Furthermore, large channels filled with water effectively served as another reference, and the obtained calibrated spectra exhibited the expected behavior (flat response at high f) and the dielectric constant of bulk water. We note that if the separation between channels is too small, calibration curves obtained above hBN spacers could, in principle, contain an electrostatic contribution coming from water itself. This would in turn lead to an underestimate of the dielectric constant of water inside the channels. To avoid such crosstalk, we have chosen the distance between the channels to be ~800 nm, large enough. Using numerical simulations, we also checked that, within our experimental accuracy (~1 $zF \cdot nm^{-1}$), the calibration curves obtained in our setup were not influenced by electrostatics coming from water inside the channels (Extended Fig. 7f).

Dielectric images and dielectric spectra acquisition. The dielectric images in Fig. 1 and Extended Data Fig. 4 were taken at the constant height z_{scan} (typically ~ 15-25 nm) from the top layer, as in Ref.[22] The images as a function of f reported in Fig. 2b were taken at both constant z_{scan} and constant y position (y-axis is along the nanochannels length). We typically recorded 25 lines at each f. Standard AFM image processing consisting of